

[Lecture Note] Measurement Errors, Reliability, Validity

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Learning Objectives

- 5-1 Recognize measurement errors and describe how to categorize them.
- 5-2 Compare and contrast the interrater reliability, test-retest reliability, and internal consistency reliability.
- 5-3 Analyze the different types of validity: face validity, content validity, construct validity, criterion validity, concurrent validity, and predictive validity.

SHORT VIDEO INTRODUCTION

<https://www.youtube.com/watch?v=EpBfGXLxQI4>

Defining Measurement Error

- Measurement error: when the data collected does not represent reality because of the way it was measured.
- Exclusive to quantitative design

Types of Measurement Errors

- Random Error: when the measurement error in the collected data is inconsistent and has no pattern.
 - When measurement error is small and random, there still may be high-quality data available.
- Systematic Error: when data collected indicates that the measure is not accurately measuring the concept.
 - Far more problematic than random error

- Can be related to the measurement being used, the way the data is being collected, or other environmental factors unrelated to the measurement.

Reliability

- Reliability in quantitative design: the consistency in a measurement; data collection is reliable if it yields the same results when used on different subjects, populations, or settings.
- Reliability in qualitative design: the study's replicability, dependability, and consistency.
- Accuracy: participants' answers about a topic and whether or not the answers are free from error.
- Reliability does not mean accuracy.

Inter-Observer or Inter-Rater Reliability

- There is more than one researcher present in data collection, and all observers keep separate ratings of the data to then compare responses to determine a degree of consistency.
- More common in qualitative studies

Test-Retest Reliability

- Associated exclusively with quantitative design
- Ensures data collection tool consistently yields the same results regardless of the passing of time.

Internal Consistency Reliability

- Often used in surveys and questionnaires and associated with quantitative designs
- Including multiple questions that measure the same variable to make sure the participant's answer was not due to error.

Validity

- Validity: the ability or potential of a data collection tool to capture and measure the construct or phenomenon that the researcher is interested in measuring.
- In quantitative design, it is tightly related to the measurement of a concept, whereas in qualitative design it relates to how a concept is being captured through texts, images, or sounds.

Face Validity

- When researchers decide to classify the measurement instrument or tool as an accurate measurement of the construct in question.
- A subjective assessment of the validity of a measurement instrument
- Does not imply the measurement is valid, but implies that for the sake of the researcher's discussion, the measurement is considered valid.
- Often used because measurement tools cannot always be confirmed as valid

Content Validity

- How well a measurement tool captures all aspects of the construct that is being tested or measured.

Construct Validity

- How well a measurement is able to truly measure the construct in question; how accurately researchers have operationalized a construct.

Criterion Validity, Concurrent Validity, and Predictive Validity

- Criterion validity: for cases when researchers are not satisfied with the way the construct has been measured and decide to create a new measurement for a construct that has already been measured.
- Criterion validity needs to establish both concurrent validity and predictive validity.
- Concurrent validity: occurs when comparing a new test with an established measurement tool and examining the matching results
- Predictive validity: occurs when a measurement test produces the same results over time.
- Difference Between Reliability and Validity of the Measurement Instrument

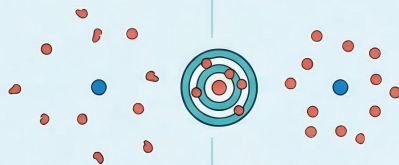
Chapter Infographic

The Pillars of Research Measurement: Error, Reliability, and Validity

Every research study faces the risk of measurement error, which occurs when collected data fails to reflect reality. To ensure study credibility, researchers must maximize Reliability (consistency) and Validity (accuracy), understanding that while a tool can be consistent without being accurate, it cannot be truly accurate without first being consistent.

Measurement Errors: Random vs. Systematic

Random Error: Inconsistent Variations



Random Error: Inconsistent Variations

Minor, patternless errors that usually allow for high-quality data analysis despite small discrepancies.

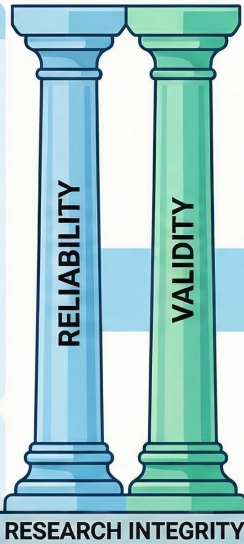
Systematic Error: Patterned Inaccuracy

Highly problematic errors where the majority of participants interpret questions incorrectly or instruments are biased.



Prevention through Pilot Testing

Conducting "trial runs" with small groups helps identify confusing questions before formal data collection.



Reliability and Validity: The Gold Standard

Reliability equals Consistency



A measure is reliable if it yields the same results across different populations and settings.

Validity equals Meaningfulness



Validity ensures the instrument is truly capturing the specific construct it intends to measure.

Validity Requires Reliability

An instrument can be reliable without being valid, but it cannot be valid without reliability.

Inter-Rater Reliability



Multiple researchers compare separate ratings for consistency.

Test-Retest Reliability



Administering the same tool to the same group at different times.

Internal Consistency Reliability



Including multiple questions within one survey that measure the same variable.

NotebookLM

NotebookLM 생성 요약 인포그래픽. 해설 영상·팟캐스트는 resource/ 참조.

Quiz

출처: 02_Assets/assessment/GE2021_TestBank.xlsx (chapter=5) · 5 문항

Q1. A carefully designed study can usually eliminate measurement errors entirely.

- A. TRUE
- B. FALSE

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정답: B

Measurement errors are present in any design; the realistic goal is to minimize them, not eliminate them.

77 **Q2. Which of the following BEST defines a measurement error?**

- 78 • **A.** A discrepancy between collected data and reality due to how the construct was measured
- 79 • **B.** A statistical mistake made during data analysis
- 80 • **C.** An ethical violation in the data collection process
- 81 • **D.** A typographical error in the final research report

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정답: **A**

A measurement error occurs when data do not represent reality because of the way the construct was measured.

83 **Q3. What is the KEY characteristic that distinguishes random error from systematic error?**

- 84 • **A.** Random error is always larger in magnitude
- 85 • **B.** Random error always invalidates the entire dataset
- 86 • **C.** Random error only occurs in qualitative studies
- 87 • **D.** Random error has no consistent pattern

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정답: **D**

Random error is inconsistent and has no pattern, while systematic error shows a consistent pattern across participants.

89 **Q4. Asking participants “How often do you feel anxious?” with options**
90 **Never/Rarely/Sometimes/Often is likely to introduce which problem?**

- 91 • **A.** Random error caused by vague qualifiers interpreted differently by participants
- 92 • **B.** Systematic error from a flawed instrument
- 93 • **C.** Predictive validity issues
- 94 • **D.** Inter-rater reliability conflicts

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정답: **A**

Words like “often” are interpreted differently by people (Barnes et al., 2003), creating random error with no consistent pattern.

Q5. A study in a Muslim and Jewish neighborhood asks “How often do you go to church?” to measure religiosity. The resulting data would suffer from:

- A. Random error due to vague terms
- B. High predictive validity
- C. Systematic error caused by a flawed measurement instrument
- D. Inter-rater reliability problems

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정답: C

The question systematically misrepresents religiosity for non-Christian populations, producing a consistent error pattern across most participants.

Reference